

# Mid/Side

A mid/side processor for width, mono bass, air on the sides and focus in the centre.



## Overview

Mid/Side splits a stereo signal into its mid (what both sides share - the centre of the image) and its side (the difference - the width). You can then set their levels separately, widen or narrow the image, keep the bass in mono, add air to the sides and presence to the centre, before it is turned back into left and right.

WIDTH keeps the loudness steady as you widen or narrow, so you hear the change in the image rather than a change in level. A vectorscope, a correlation meter and mid / side energy bars show what is happening.

It can also take in or send out mid/side signals directly, as an M/S encoder or decoder.

## Reach for it for

- Widening or narrowing a mix or a stereo bus
- Keeping the low end in mono for a solid, cut-friendly bass
- Adding air to the edges without making the centre harsh

- Bringing a vocal forward in the centre (FOCUS)
- Checking a mix in mid or side solo
- Encoding or decoding M/S recordings

## Controls

### Display

|                    |                                                                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vectorscope</b> | The output as a cloud of dots: a vertical line is mono, a wide cloud is wide stereo, and a horizontal spread means the sides are out of phase.                          |
| <b>CORRELATION</b> | From -1 to +1. Near +1 the sides are mostly the same (mono-safe); around 0 is very wide; below 0 (the bar turns red) the sides are out of phase and may cancel in mono. |
| <b>MID / SIDE</b>  | The energy of the mid and side signals.                                                                                                                                 |
| <b>INPUT</b>       | A VU meter showing the input level (0 VU = -18 dBFS).                                                                                                                   |

### Header

|                            |                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>STEREO / MID / SIDE</b> | What you hear. STEREO is the normal output; MID plays the mid signal alone and SIDE plays the side signal alone (in both ears). MID and SIDE are for listening only - switch back to STEREO before you bounce. |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### M/S

|                          |                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>IN L/R / IN M/S</b>   | The input format. L/R: a normal stereo input. M/S: the input is already mid (left channel) and side (right channel). |
| <b>OUT L/R / OUT M/S</b> | The output format. L/R: a normal stereo output. M/S: sends out mid on the left and side on the right, for encoding.  |
| <b>MID</b>               | The level of the mid signal, -12 to +12 dB.                                                                          |
| <b>SIDE</b>              | The level of the side signal, -12 to +12 dB.                                                                         |

|                  |                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>WIDTH</b>     | 0 to 200% (100% leaves the image as it is). 0% is mono; above 100% is wider. The overall loudness is kept steady as you turn it.      |
| <b>BASS MONO</b> | Removes the side signal below this frequency (24 dB per octave), up to 300 Hz, so the low end sits in mono. Below 20 Hz it reads OFF. |
| <b>AIR</b>       | A high shelf at 8 kHz on the side signal only, 0 to +9 dB: an open, airy top to the edges without brightening the centre.             |
| <b>FOCUS</b>     | A presence lift at 2.5 kHz on the mid only, 0 to +6 dB: brings vocals and other centre sounds forward.                                |

## DRIFT

|              |                  |
|--------------|------------------|
| <b>DRIFT</b> | See Drift below. |
|--------------|------------------|

## IN / OUT

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TRIM</b>               | Input level before anything else, from -24 to +12 dB. A single click on the knob switches GUARD on or off (on by default); its light at the knob's upper right is lit while GUARD is on and glows brighter while it is catching peaks. GUARD gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. |
| <b>SAT</b>                | Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.                                                                                                                                                                                                                                                                                 |
| <b>OFF / CLIP / LIMIT</b> | The final stage: nothing, a soft clipper (no latency, the default), or a lookahead limiter (3 ms latency, compensated by REAPER). The CLIP key flashes red while the output is being clipped.                                                                                                                                                                                                                                   |

## MASTER

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| <b>BALANCE</b>    | Left / right balance, -100 to +100%.                             |
| <b>OUTPUT</b>     | The output level, -24 to +12 dB.                                 |
| <b>OUT meters</b> | Output level, left and right. Click them to reset the peak hold. |

## The header

|                     |                                                                                                                                                                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Theme picker</b> | Changes the panel's colour theme.                                                                                                                                                                                                                                                                                    |
| <b>S / M / L</b>    | Panel size: small, medium or large. The panel also scales to fit the plugin window.                                                                                                                                                                                                                                  |
| <b>HW</b>           | Switches between the hardware-style panel and the flat look.                                                                                                                                                                                                                                                         |
| <b>PRESETS</b>      | Opens the list of factory presets (or click the preset display). The ◀ / ▶ keys step through them. Choosing one sets the M/S section and the drift amounts; the listen mode, BALANCE, OUTPUT, the DRIFT dial and the input / output settings are left as they are. The last entry resets everything to the defaults. |
| <b>?</b>            | Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.                                                                                                                                                                                           |

## Drift

DRIFT (0 - 100%, 50% by default) brings slow, random movement: the width breathes, the AIR shelf wanders and the left / right balance leans a little. 50% is the standard amount, 100% twice that, 0 none; the wander also speeds up as you turn it up. Some presets set their own amounts for the three movements (the Mono Bass, Narrow & Tight and M/S presets switch them off).

## Presets

|                     |                                 |
|---------------------|---------------------------------|
| <b>Subtle Widen</b> | 120% width with a touch of air. |
|---------------------|---------------------------------|

|                              |                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------|
| <b>Wide Master</b>           | 140% width, a little more side level and air, bass in mono below 120 Hz.            |
| <b>Mono Bass</b>             | No width change: just the bass in mono below 150 Hz, no drift.                      |
| <b>Vocal Focus</b>           | Slightly narrower (90%), the mid up 1 dB and 3 dB of FOCUS, mono bass below 100 Hz. |
| <b>Super Wide</b>            | 175% width, more side level, 4 dB of air, mono below 200 Hz, more drift.            |
| <b>Narrow &amp; Tight</b>    | 70% width and the sides down 2 dB, mono below 150 Hz, no drift.                     |
| <b>M/S Decode</b>            | Takes a mid/side input and turns it into left and right.                            |
| <b>M/S Encode</b>            | Takes a left/right input and sends out mid (left) and side (right).                 |
| <b>Reset all to defaults</b> | Every control back to its starting value.                                           |

## Tips

- Use SIDE listen to hear what WIDTH and AIR are really adding; if it sounds harsh or hollow, back off.
- Keep an eye on CORRELATION while widening: if it heads below 0, the mix will lose energy in mono.
- BASS MONO at 100-150 Hz tightens the low end of almost any mix without changing the image above it.
- FOCUS is subtler than an EQ on the whole mix: it lifts the centre only, so the sides stay as they are.

File: gfx-midside.jsfx